Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To Eliminate Failures of Cleco DC Tools Breakdown / Failure Concern ".

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**02.05.19**	**02.05.2019**
Completion Date	**30.12.19**	**20.12.19**

Current Status: **Open**

Cross Functional Team

Sr. No	Description	Name	Designation	Dept.	Phone
1	**Team Leader**	Jitendra Bhamare	Asst. Manager	Maintenance	8975856767
2	**Mentor**	Tushar Patil	Sr. Manager	Maintenance	9922409136
3	**Member 1**	Hari Nagar	Manager	Maintenance	9881786767
4	**Member 2**	Sandip Ghotekar	Asst. Manager	Maintenance	9850848696
5	**Member 3**	Ajit Chaudhary	Asst. Manager	Production	7588616388
6	**Member 4**	Shreedutta Desai	Service Engg.	OEM Supplier	8879008583

Step1. Select Subject

Title :

To eliminate Cleco DC tool Failure.

Brief Description :

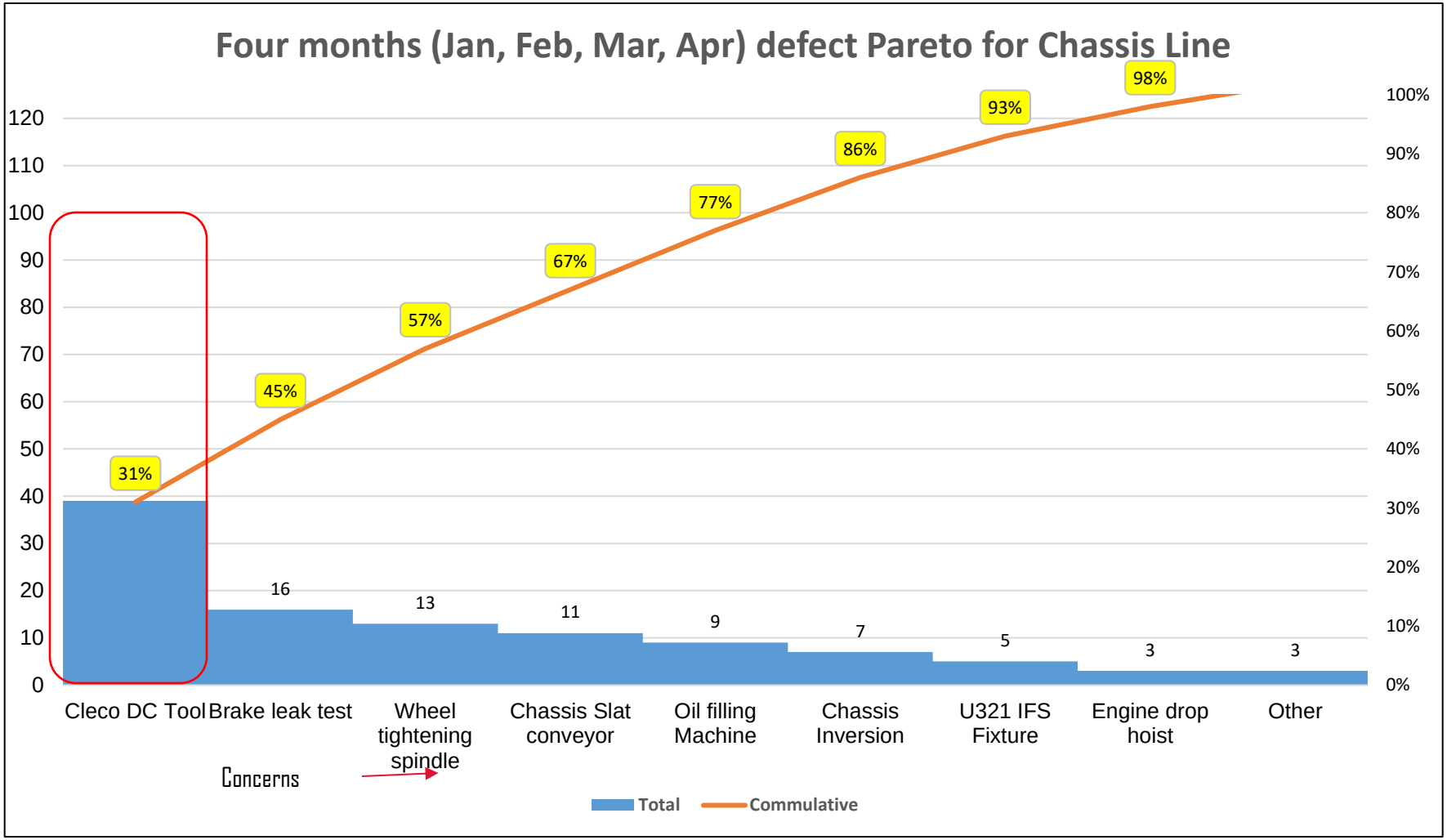
WHAT	W105 Scorpio / Marazzo Vehicle bolt tightening Cleco DC tool failure
WHERE	1. At Final line in Multimodel TCF shop 2. Body drop stage
WHEN	Since Jan-19 month
WHO	By stage operator
WHY	1. Leading to line loss 2. High rework time 3. Quality concern, bolt loose.
HOW	Avg. 6 cases per day

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Inference – Cleco DC Tool Failure is the top defect at Chassis line related to Maint.

2. Justify the choice

COST OF POOR QUALITY- REWORK

Importance & Consequences of This Concern -

- Major rework time for this issue.
- FRC loss at Final line

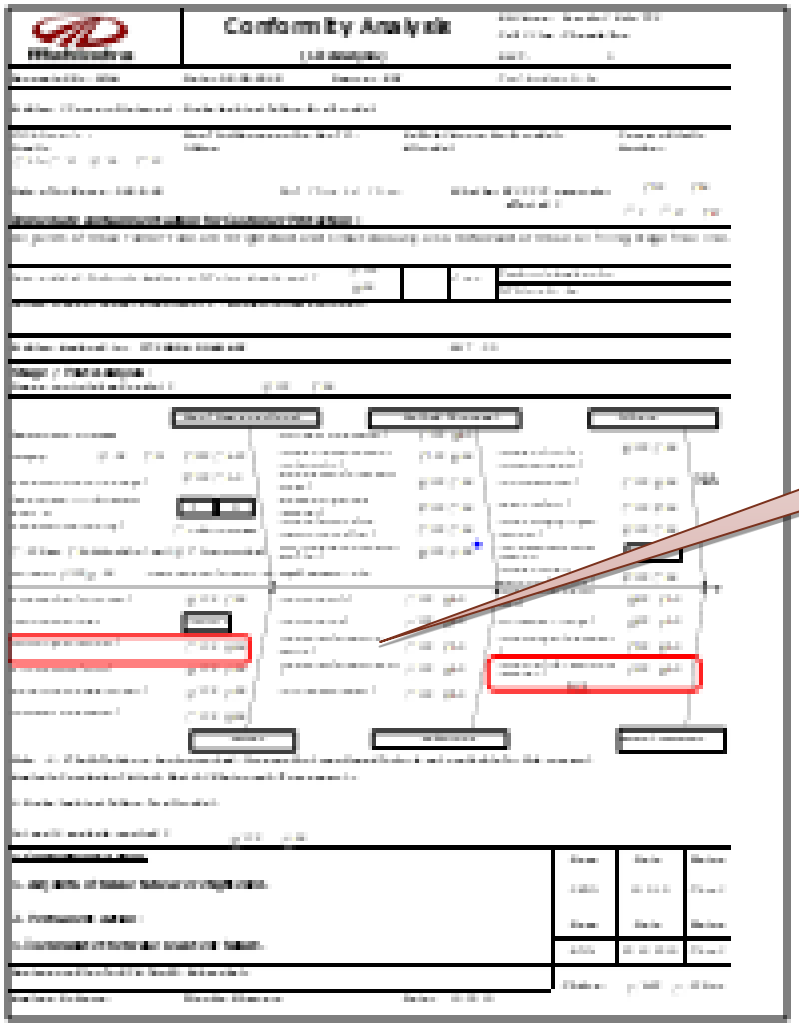
Rework Cost Calculation -

Rework cost estimation for Cleco DC Tool Failure issues		
Description	Units	
Rework time(Approx)	Min	240
Avg Concerns/Day	Nos	0.13
Total Required Rework time per day	Min	31.2
Effective working time	Min	465
Operator required/day	Nos	0.06
Approx Operator cost/ day	Rs	1360
Operator cost / concern	Rs	75.2
Avg no. of working days	days	25
Operator cost /month	Rs	1880
Operator cost/year	**Rs**	22560

2.b Immediate Containment Action

- Awareness given to stage operators for less torque or Bolt loose concern.
- Torque wrench given to stage associate to ensure the tightening torque of the joint
- Awareness given to Final line buy off stage.

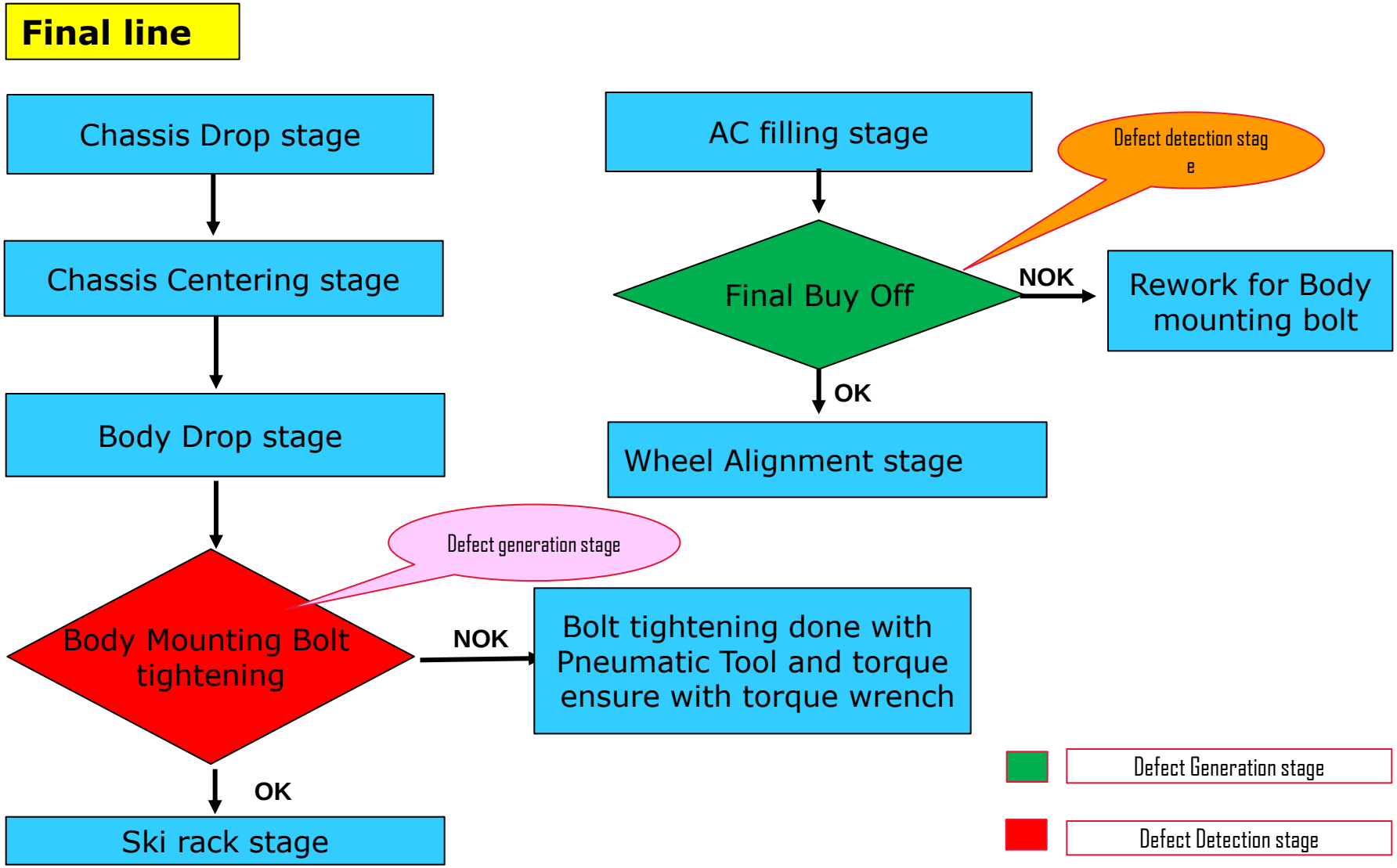
Step 3. Understanding Current Situation



Inference from L0,

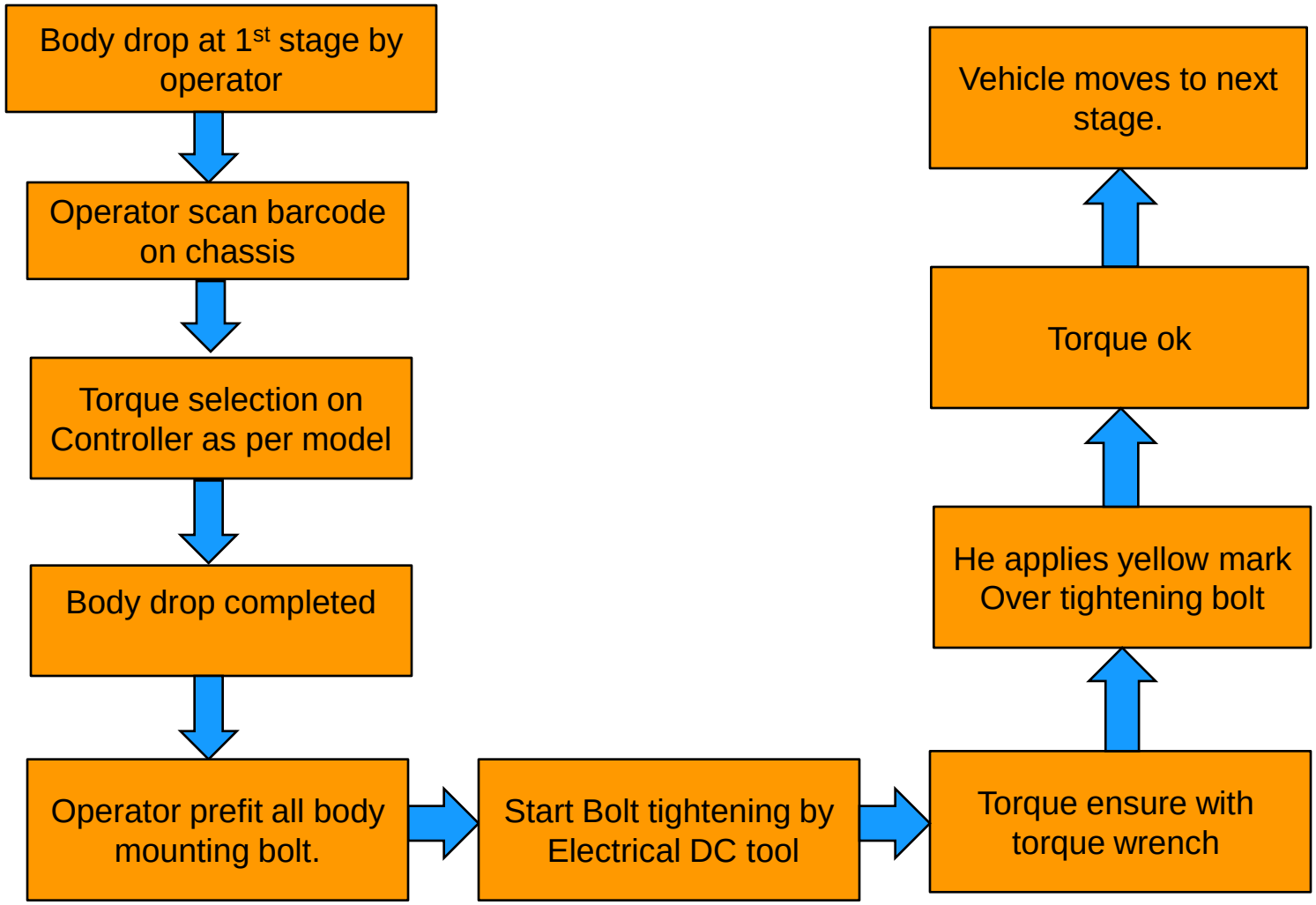
4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L4
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Cleco electrical DC Tool .
Material	Handling and shelf life of part adhered

3.Understand The Current Situation

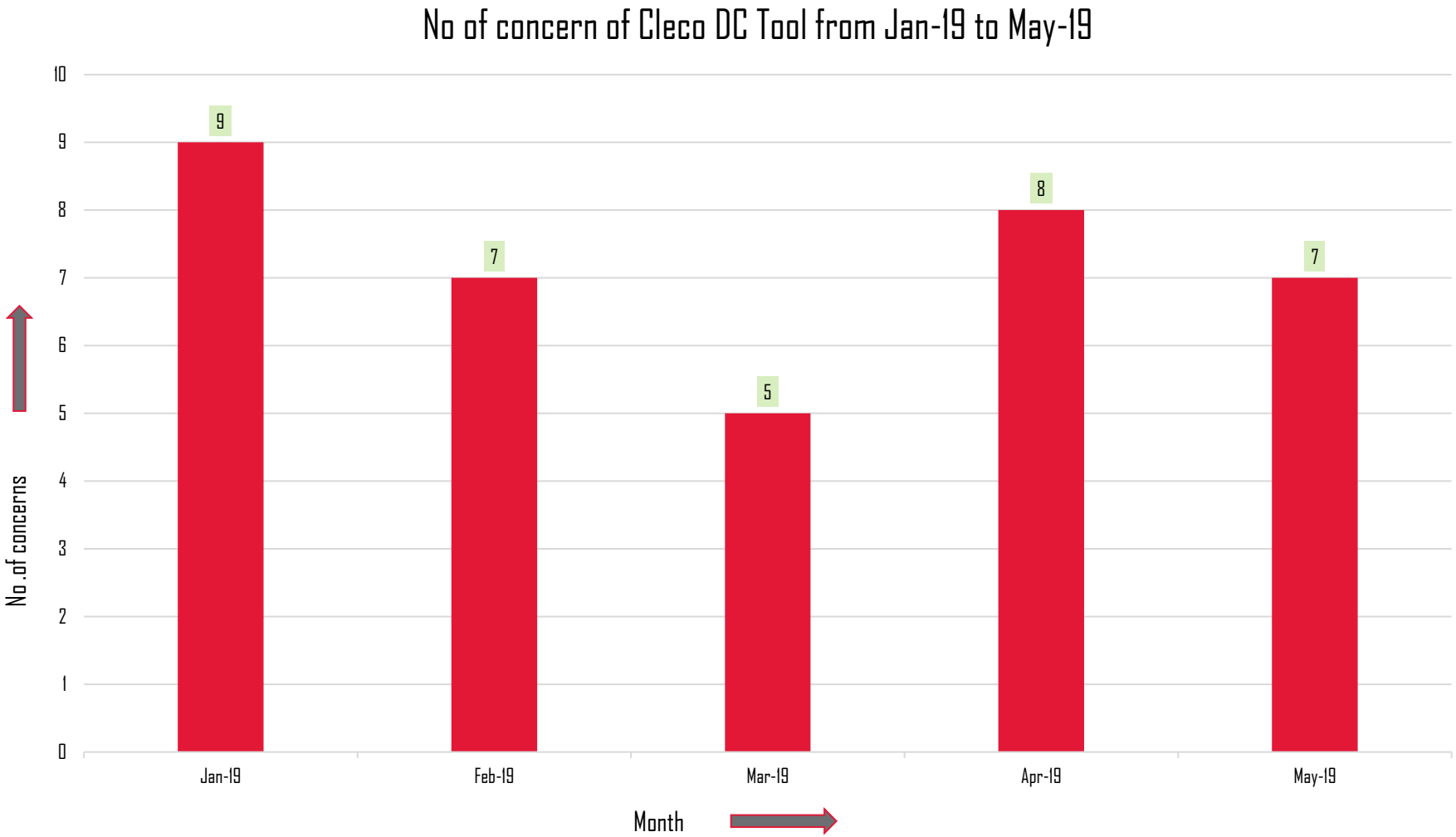


3. Understand The Current Situation

Actual Flow of Cycle :



3. Understand the current situation



Inference – Avg.6 defect reported at chassis line for this defect.

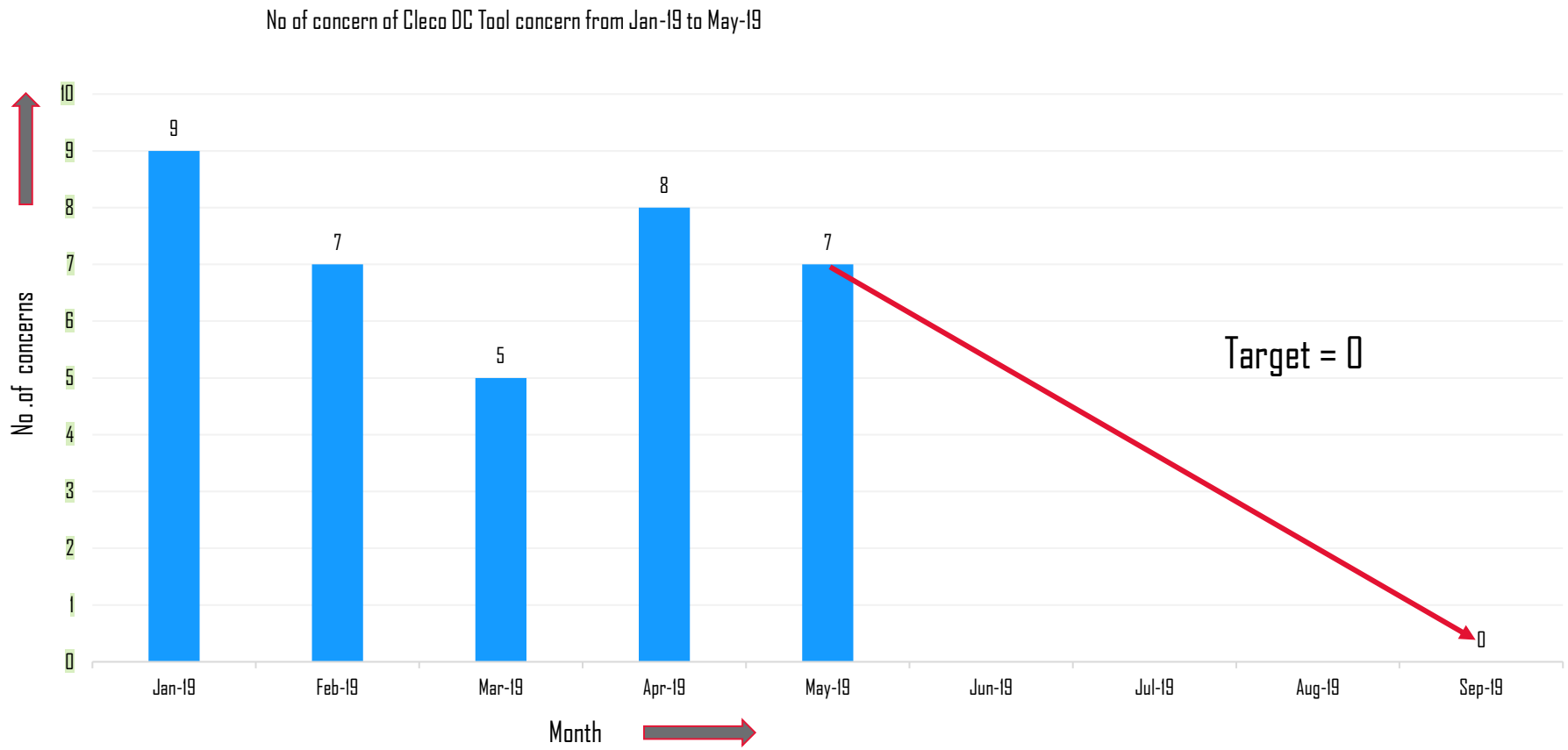
Step 3. Understanding Current Situation

Good & Not so good observations

- Issue reported in both the shifts
- Repair time is around 53 minutes per vehicle after roll down.
- Issue reported Buy-off.
- There was level 3 operators for Body mounting bolt tightening.
- All the vehicles having this issue.

4. Select targets

Present Level:	36 cases reported at FAI based on data of last 05 months
Target :	**Concern elimination by end of Sept 2019**



5. Analysis

List of Possible causes after brainstorming among team members

Sr. No	Possible Cause
1	Barcode scanning issue
2	Ele. DC tool communication lost
3	Communication/supply cable faulty
4	Controller version error
5	PLC fault
6	Controller display problem
7	DC tool PY issue
8	DC tool trigger issue
9	Balancer issue
10	Ele. DC tool version error
11	Tool transducer fault
12	PLC fault
13	Ele. DC tool angle head problem
14	New Operator deployed on stage
15	Stage operator skill level
16	Cycle selection wrong(Marazzo/W105)
17	SOS not followed
18	Bolt mounting hole thread damage
19	Body mounting bolt harness issue
20	Ele tool socket worn out

Step 5. Analysis

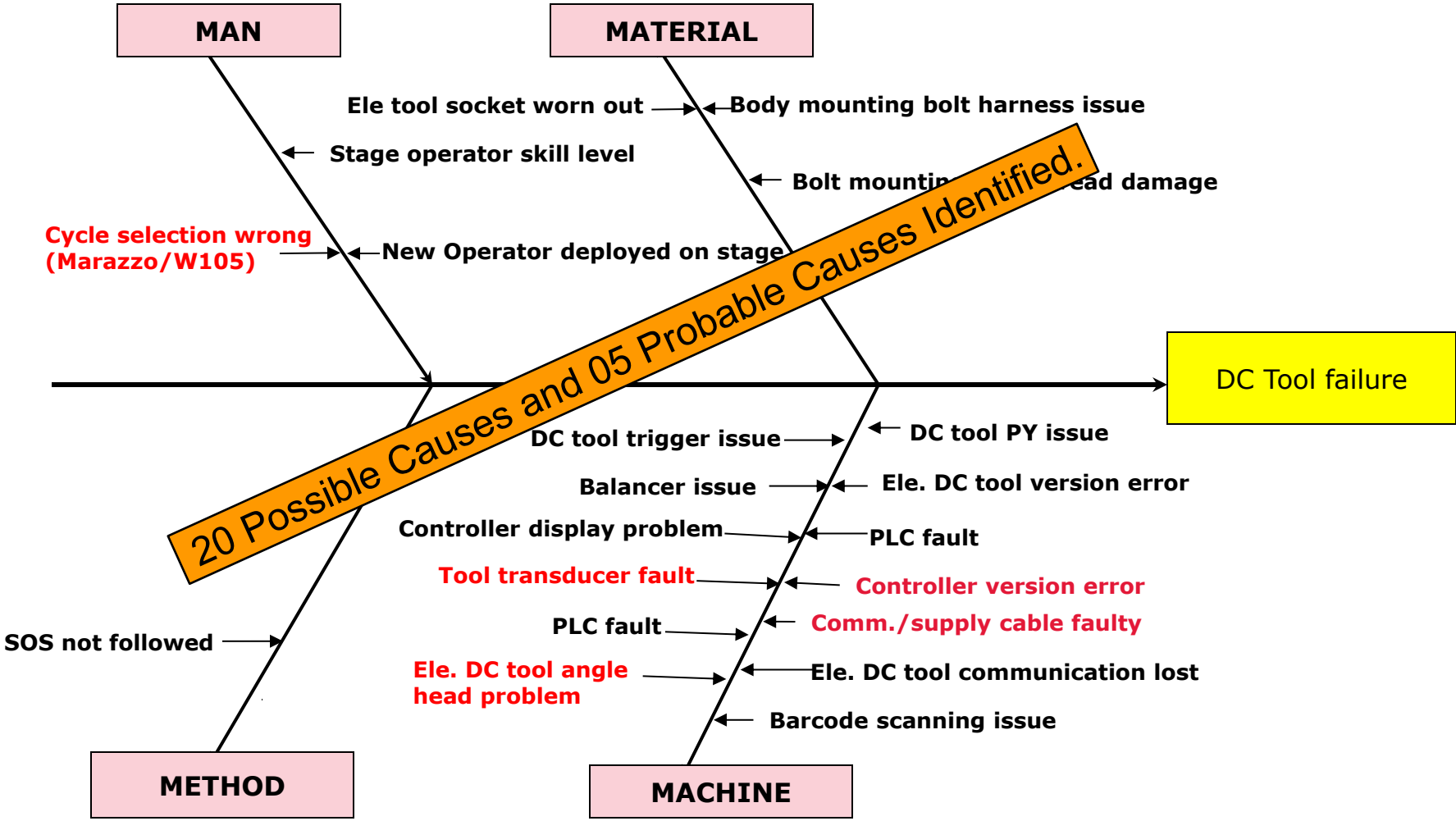
List of Probable causes identification in Brain storming session

Sr. No	Probable Cause	Jitendra	Hari	Tushar	Ajit	Shreedutt	Sandip	Total
1	Barcode scanning issue	1	1	1	1	1	1	6
2	Ele. DC tool communication lost	1	3	3	1	3	3	14
3	Communication/supply cable faulty	9	9	3	3	9	9	42
4	Controller version error	9	9	9	3	3	9	42
5	PLC fault	3	3	3	3	1	3	16
6	Controller display problem	1	3	3	3	1	3	14
7	DC tool PY issue	1	3	1	3	3	1	12
8	DC tool trigger issue	1	1	3	1	3	3	12
9	Balancer issue	1	3	3	9	1	1	18
10	Ele. DC tool version error	3	3	9	9	1	3	28
11	Tool transducer fault	9	9	9	3	9	9	48
12	PLC fault	1	1	1	3	3	9	18
13	Ele. DC tool angle head problem	3	9	9	9	3	9	42
14	New Operator deployed on stage	1	3	1	1	3	3	12
15	Stage operator skill level	1	1	3	3	3	9	20
16	Cycle selection wrong(Marazzo/W105)	9	9	3	3	9	9	42
17	SOS not followed	3	1	3	3	3	1	14
18	Bolt mounting hole thread damage	1	1	3	1	1	3	10
19	Body mounting bolt harness issue	3	3	1	1	1	3	12
20	Ele tool socket worn out	1	3	3	3	3	1	14

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 42.

5. Analysis

5.1 Identify Possible Causes: Fish Bone Diagram



Step 5. Analysis

List of causes not considered as probable causes and reasons for not considered

Sr No	Not considered causes	Reasons
1	SOS not followed	SOS given and display on stage
2	New Operator deployed on stage	Operator is level 3
3	Balancer issue	Checked with weighting scale
4	PLC fault.	No PLC fault on PLC
5	DC tool PY issue	PLC program checked PY is working
6	Ele. DC tool socket problem.	Pushbutton marking as per model
7	Tool trigger issue	Magnetic switch is provided
8	Barcode scanning issue	Scanner communication checked with Hyperterminal
9	Controller tripping problem.	All wiring checked & verified
10	Ele. DC tool communication lost	Communication checked with Apex software
11	Stage operator skill level	Operator is level 3
12	Controller display problem	Torque checked with apex software
13	Ele. DC tool version error	DC tool version verified with OEM
14	Bolt mounting hole thread damage	All parts are as per quality norms
15	Body mounting bolt harness issue	No Bolt broken issue

Step 5. Analysis

List of causes considered as probable causes and testing method :-

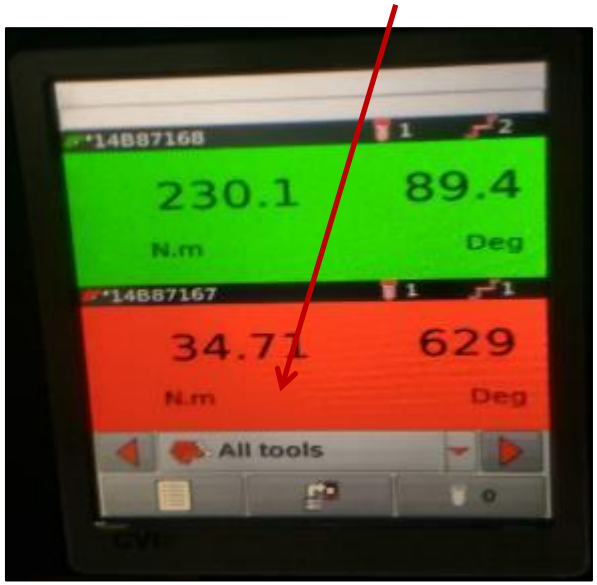
Sr No	Possible causes	Testing Method
1	Ele. DC tool angle head problem	Acta tool measurement unit.
2	Communication/supply cable faulty.	Visual inspection of communication cable connector though Apex software.
3	Cycle selection wrong(Marazzo/W105)	Visual Inspection
4	Ele. DC tool version error.	Apex software communication
5	Tool transducer problem.	Visual Inspection

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Ele. DC tool angle head problem. (Bevel gear)	Acta tool measurement unit.	▪ During tightening DC tool not deliver full torque 24 No's cycles checked 8 found not ok. ▪ Abnormal sound came during tightening. ▪ Torque reading shows on controller display.	**Hypothesis valid**

Angle head assembly



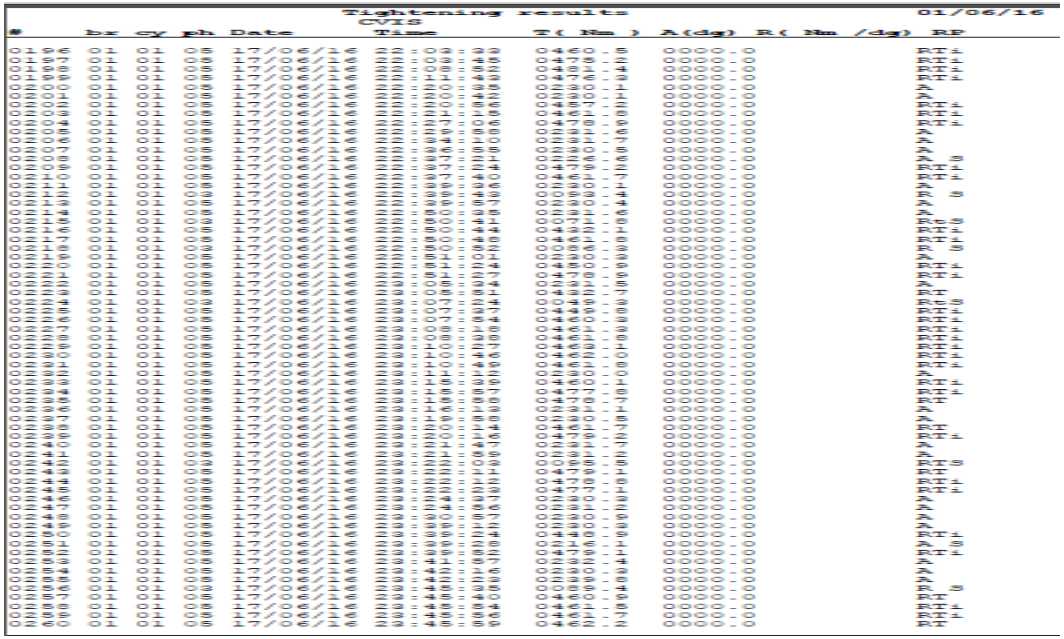
Sr. No	Status	ID	Tool	Torque	Angle	Stop source	Date	Result	TU
1	OK	622	1	230.4	107	Target reached	12-May-16	1	2
2	NOK	621	2	34.99	1071	Trigger released	12-May-16	1	2
3	OK	620	1	230.7	86.1	Target reached	12-May-16	1	2
4	OK	619	2	230.6	82.8	Target reached	12-May-16	1	2
5	NOK	618	1	37.84	1156	Trigger released	12-May-16	1	2
6	OK	617	2	230.4	103	Target reached	12-May-16	1	2
7	OK	616	1	230.5	105	Target reached	12-May-16	1	2
8	NOK	615	2	59.85	677	Trigger released	12-May-16	1	2
9	NOK	614	1	23.92	610	Trigger released	12-May-16	1	2
10	OK	613	2	230.4	125	Target reached	12-May-16	1	2
11	OK	612	1	230.3	132	Target reached	12-May-16	1	2
12	NOK	611	2	53.75	1256	Trigger released	12-May-16	1	2
13	OK	610	1	230.6	109	Target reached	12-May-16	1	2
14	OK	609	2	230.9	106	Target reached	12-May-16	1	2
15	NOK	608	1	47	1192	Trigger released	12-May-16	1	2
16	OK	607	2	230.8	107	Target reached	12-May-16	1	2
17	OK	606	1	230.6	154	Target reached	12-May-16	1	2
18	NOK	605	2	62.06	1333	Trigger released	12-May-16	1	2
19	OK	604	1	230.3	87.2	Target reached	12-May-16	1	2
20	OK	603	2	230.1	108	Target reached	12-May-16	1	2
21	OK	602	1	230.1	115	Target reached	12-May-16	1	2
22	OK	601	2	231	86.8	Target reached	12-May-16	1	2
23	NOK	600	1	40.05	1152	Trigger released	12-May-16	1	2
24	OK	599	2	230.7	130	Target reached	12-May-16	1	2

DC tool cycles observed – 24 checked , 8 point not ok.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Communication/supply cable connector faulty	Visual inspection of communication cable connector though Apex software.	▪ Comm. Pin head found burnt. ▪ Cable lock wear out, not locking properly.	**Hypothesis Valid**



Tightening results 01/06/16

CVIS

#	bx	cy	ph	Date	Time	T(Nm)	A(dg)	R(Nm /dg)	RP
01967	01	01	05	17/06/16	22:03:33	0460.5	00000.0		RT1
01968	01	01	05	17/06/16	22:03:45	0475.4	00000.0		RT1
01969	01	01	05	17/06/16	22:08:52	0481.4	00000.0		RT1
01970	01	01	05	17/06/16	22:15:13	0476.3	00000.0		RT1
02000	01	01	05	17/06/16	22:20:35	0530.1	00000.0		A
02001	01	01	05	17/06/16	22:20:42	0530.1	00000.0		A
02002	01	01	05	17/06/16	22:20:42	0475.2	00000.0		RT1
02003	01	01	05	17/06/16	22:21:15	0461.8	00000.0		RT1
02004	01	01	05	17/06/16	22:27:05	0478.5	00000.0		RT1
02005	01	01	05	17/06/16	22:29:06	0531.4	00000.0		A
02006	01	01	05	17/06/16	22:34:10	0531.7	00000.0		A
02007	01	01	05	17/06/16	22:36:15	0530.5	00000.0		A
02008	01	01	05	17/06/16	22:37:21	0526.4	00000.0		A S
02009	01	01	05	17/06/16	22:37:24	0475.5	00000.0		RT1
02010	01	01	05	17/06/16	22:37:40	0461.7	00000.0		RT1
02011	01	01	05	17/06/16	22:38:36	0530.1	00000.0		A
02012	01	01	05	17/06/16	22:38:43	0530.4	00000.0		A S
02013	01	01	05	17/06/16	22:40:37	0530.4	00000.0		A
02014	01	01	05	17/06/16	22:40:35	0531.4	00000.0		A
02015	01	01	05	17/06/16	22:40:44	0475.1	00000.0		RT S
02016	01	01	05	17/06/16	22:40:44	0532.1	00000.0		RT1
02017	01	01	05	17/06/16	22:40:48	0461.8	00000.0		RT1
02018	01	01	05	17/06/16	22:50:17	0476.9	00000.0		A S
02019	01	01	05	17/06/16	22:51:01	0530.3	00000.0		A
02020	01	01	05	17/06/16	22:51:24	0458.9	00000.0		RT1
02021	01	01	05	17/06/16	22:51:37	0478.9	00000.0		RT1
02022	01	01	05	17/06/16	23:05:34	0531.5	00000.0		A
02023	01	01	05	17/06/16	23:05:51	0532.7	00000.0		RT
02024	01	01	05	17/06/16	23:07:14	0478.5	00000.0		RT S
02025	01	01	05	17/06/16	23:07:37	0460.8	00000.0		RT1
02026	01	01	05	17/06/16	23:07:54	0460.3	00000.0		RT1
02027	01	01	05	17/06/16	23:08:18	0461.3	00000.0		RT1
02028	01	01	05	17/06/16	23:08:38	0461.8	00000.0		RT1
02029	01	01	05	17/06/16	23:10:46	0462.0	00000.0		RT1
02030	01	01	05	17/06/16	23:10:48	0461.8	00000.0		RT1
02031	01	01	05	17/06/16	23:15:39	0460.1	00000.0		A
02032	01	01	05	17/06/16	23:15:57	0477.8	00000.0		RT1
02033	01	01	05	17/06/16	23:16:13	0531.1	00000.0		RT
02034	01	01	05	17/06/16	23:16:18	0530.5	00000.0		A
02035	01	01	05	17/06/16	23:20:16	0476.2	00000.0		A
02036	01	01	05	17/06/16	23:21:40	0461.4	00000.0		RT
02037	01	01	05	17/06/16	23:21:49	0478.2	00000.0		RT1
02038	01	01	05	17/06/16	23:21:55	0531.2	00000.0		A
02039	01	01	05	17/06/16	23:22:02	0526.5	00000.0		RT S
02040	01	01	05	17/06/16	23:22:17	0475.1	00000.0		RT
02041	01	01	05	17/06/16	23:22:18	0478.8	00000.0		RT1
02042	01	01	05	17/06/16	23:22:29	0477.1	00000.0		RT1
02043	01	01	05	17/06/16	23:24:56	0531.4	00000.0		A
02044	01	01	05	17/06/16	23:30:57	0530.5	00000.0		A
02045	01	01	05	17/06/16	23:35:24	0478.9	00000.0		RT1
02046	01	01	05	17/06/16	23:35:26	0478.1	00000.0		A S
02047	01	01	05	17/06/16	23:35:27	0478.1	00000.0		RT1
02048	01	01	05	17/06/16	23:41:57	0532.4	00000.0		A
02049	01	01	05	17/06/16	23:42:16	0532.9	00000.0		A
02050	01	01	05	17/06/16	23:42:18	0526.4	00000.0		A S
02051	01	01	05	17/06/16	23:45:35	0526.4	00000.0		RT
02052	01	01	05	17/06/16	23:45:47	0461.5	00000.0		RT1
02053	01	01	05	17/06/16	23:45:54	0461.7	00000.0		RT1
02054	01	01	05	17/06/16	23:45:56	0462.2	00000.0		RT

DC tool cycles observed – 64 checked , 41 point not ok.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Ele. DC tool version error	Visual Inspection on controller display	▪ Ele. DC tool parameter must read in its memory. ▪ Tool memory & controller updated. ▪ Controller has detected parameter which are of same from the ones saved controller memory.	**Hypothesis invalid**

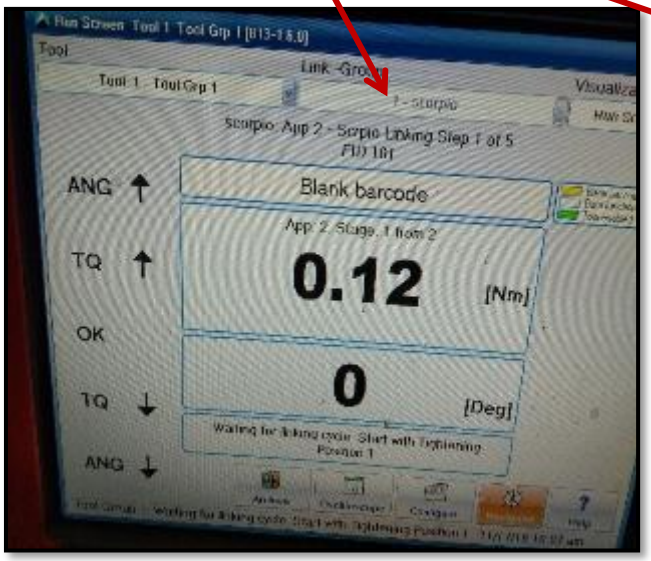
DC tool Controller display

Tool parameters found ok

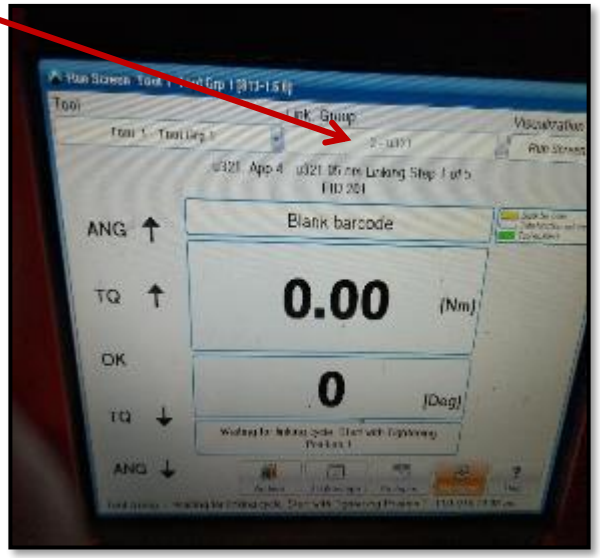
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Cycle selection wrong(Marazzo/Scorpio)	Visual Inspection	1. Cycle selection done by scanning barcode on build sheet by stage associate according to selection torque selection happen. 2. Red stack light indication is that cycle selected	**Hypothesis invalid**



Scorpio cycle selected



Marazzo cycle selected

Stack light indication that cycle selection done

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	DC tool transducer problem	Visual inspection	• During tightening DC tool torque value variation on controller display for some cycle . • Torque variation observed when transducer wire pinch.	**Hypothesis valid**

Transducer wire pinch due to rubber sleeve damaged

Sr. No	Status	ID	Tool	Torque	Angle	Stop source	Date	Result	TU
1	OK	8600	1	230.9	107	Target reached	3-Jul-16	1	2
2	NOK	8599	2	37.09	107	Trigger released	3-Jul-16	1	2
3	NOK	8598	1	37.09	86.1	Trigger released	3-Jul-16	1	2
4	NOK	8597	2	108.3	82.8	Trigger released	3-Jul-16	1	2
5	OK	8596	1	230.8	115	Target reached	3-Jul-16	1	2
6	OK	8595	2	230	103	Target reached	3-Jul-16	1	2
7	OK	8594	1	230.8	105	Target reached	3-Jul-16	1	2
8	NOK	8593	2	0	153	Trigger released	3-Jul-16	1	2
9	OK	8592	1	230.7	132	Target reached	3-Jul-16	1	2
10	NOK	8591	2	40.2	125	Trigger released	3-Jul-16	1	2
11	NOK	8590	1	40.9	132	Trigger released	3-Jul-16	1	2
12	OK	8589	2	230.1	105	Target reached	3-Jul-16	1	2
13	NOK	8588	1	37.2	109	Trigger released	3-Jul-16	1	2
14	NOK	8587	2	108.3	106	Trigger released	3-Jul-16	1	2
15	NOK	8586	1	109.2	119	Trigger released	3-Jul-16	1	2
16	NOK	8585	2	202.3	107	Trigger released	3-Jul-16	1	2
17	OK	8584	1	230.7	154	Target reached	3-Jul-16	1	2
18	OK	8583	2	230.5	133	Target reached	3-Jul-16	1	2
19	NOK	8582	1	221.8	87.2	Trigger released	3-Jul-16	1	2
20	NOK	8581	2	210.3	108	Trigger released	3-Jul-16	1	2
21	OK	8580	1	230.3	115	Target reached	3-Jul-16	1	2
22	NOK	8579	2	39.8	86.8	Trigger released	3-Jul-16	1	2
23	OK	8578	1	230.8	115	Target reached	3-Jul-16	1	2
24	NOK	8577	2	109.3	130	Trigger released	3-Jul-16	1	2

GR tool cycles observed – 24 checked , 14 point not ok.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr No	Possible causes	Testing Method	Conclusion
1	Ele. DC tool angle head problem.	Acta tool measurement unit	Hypothesis valid
2	Communication/supply cable faulty	Visual inspection of communication cable connector though Apex software.	Hypothesis valid
3	Ele. DC tool version error.	Visual Inspection	Hypothesis invalid
4	Cycle selection wrong(Marazzo/W105)	Visual Inspection	Hypothesis invalid
5	DC tool transducer problem	Apex software & Visual inspection	Hypothesis valid

TEST OF HYPOTHESIS

5.3 Identify the root causes:

Sr. No.	Probable cause	WHY?	WHY?	WHY?	Action
1	Ele. DC tool angle head problem.	Angle head wear out.	Angle head Bevel gear damaged.	DC tool reached to its maximum cycle limit.	1.DC tool angle head to be replaced. 2. Angle head extension to be provided.
2	Communication/ supply cable faulty.	Cable pin short.	Communication cable pin loosely connected.	Communication cable contact pin holder assembly loose.	Replace the cable.
3	GR tool transducer problem.	Transducer wire pinch.	Transducer wire protection rubber sleeve wear out.	Rubber sleeve of small size & thin. No protection to rubber sleeve.	protection to be provide

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1.	DC Tool angle head problem.	1.DC tool angle head to be replaced. 2. Angle head extension to be provided.	09.08.2019	Jitendra	Closed
2.	DC tool transducer problem.	Replace the cable.	07.09.2019	Jitendra	Closed
3	Communication/ supply cable connector faulty.	protection to be provide	03.08.2019	Jitendra	Closed

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	DC Tool angle head problem.	1.DC tool angle head to be replaced. 2. Angle head extension to be provided.	09.08.2019	Jitendra	Closed

tool angle head to be replaced

New extension implemented on gun

New Extension

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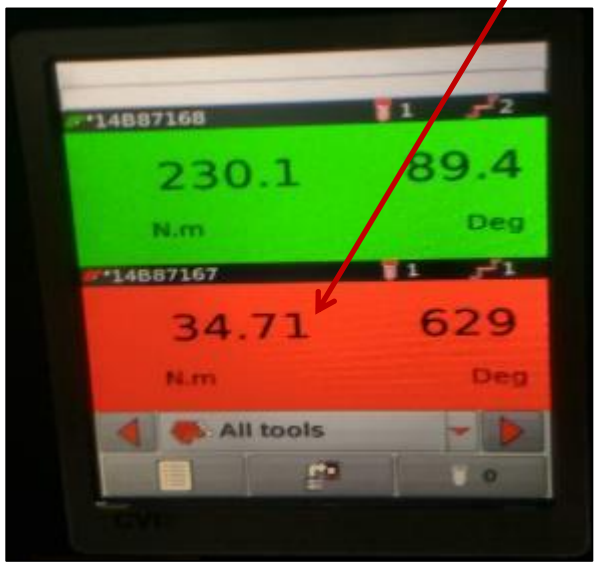
6. Implement the corrective measures

6.1 Actions Decided

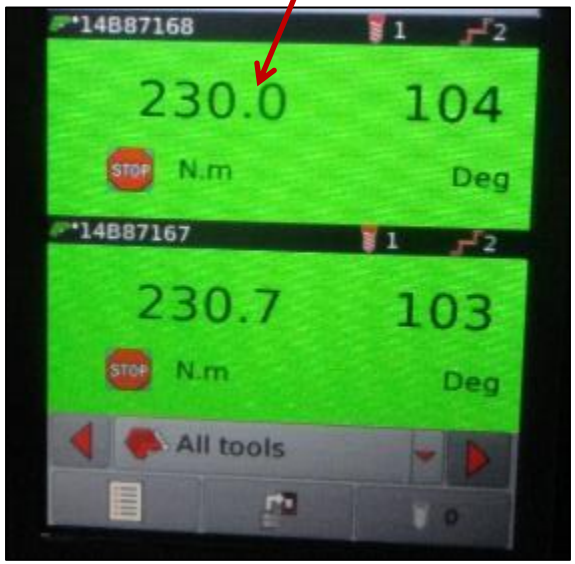
S N	Action Taken	Testing Method	Results Before Action	Results After Action
1)	Angle head replaced & angle head extension provided.	On controller display reading shows the actual torque value which is also verified on Acta tool torque measurement unit shows equal value.	DC tool Torque value below 230NM	Tool torque value reach to 230NM required torque value.

Sr. No	Status	ID	Tool	Torque	Angle	Stop source	Date	Result	TU
1	OK	15253	1	40.25	107	Trigger released	10-Aug-16	1	2
2	OK	15252	2	230.9	107	Target reached	10-Aug-16	1	2
3	OK	15251	1	230.7	86.1	Target reached	10-Aug-16	1	2
4	OK	15250	2	37.09	82.8	Trigger released	10-Aug-16	1	2
5	OK	15249	1	230.8	115	Target reached	10-Aug-16	1	2
6	OK	15248	2	230	103	Target reached	10-Aug-16	1	2
7	OK	15247	1	230.8	105	Target reached	10-Aug-16	1	2
8	OK	15246	2	230.2	153	Target reached	10-Aug-16	1	2
9	OK	15245	1	230.7	132	Target reached	10-Aug-16	1	2
10	OK	15244	2	230.4	125	Target reached	10-Aug-16	1	2
11	OK	15243	1	230.3	132	Target reached	10-Aug-16	1	2
12	OK	15242	2	230.1	105	Target reached	10-Aug-16	1	2
13	OK	15241	1	230.8	109	Target reached	10-Aug-16	1	2
14	OK	15240	2	230.7	106	Target reached	10-Aug-16	1	2
15	OK	15239	1	230.9	119	Target reached	10-Aug-16	1	2
16	OK	15238	2	230	107	Target reached	10-Aug-16	1	2
17	OK	15237	1	230.7	154	Target reached	10-Aug-16	1	2
18	OK	15236	2	230.5	133	Target reached	10-Aug-16	1	2
19	OK	15235	1	230.3	87.2	Target reached	10-Aug-16	1	2
20	OK	15234	2	230.1	108	Target reached	10-Aug-16	1	2
21	OK	15233	1	230.3	115	Target reached	10-Aug-16	1	2
22	OK	15232	2	230.4	86.8	Target reached	10-Aug-16	1	2
23	OK	15231	1	230.8	115	Target reached	10-Aug-16	1	2
24	OK	15230	2	109.3	130	Trigger released	10-Aug-16	1	2

DC tool cycles observed – 24 checked , All points ok.



Before



After

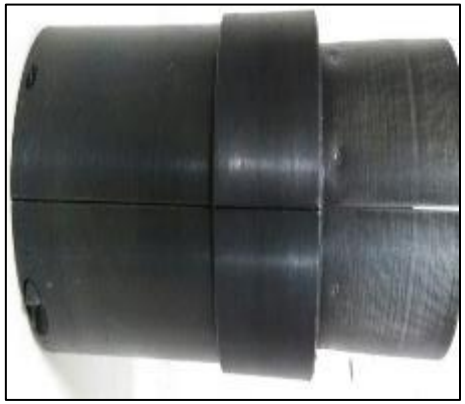
6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2	DC tool transducer problem.	DC tool transducer wire protection sleeve provided.	7.09.2019	Jitendra	Closed

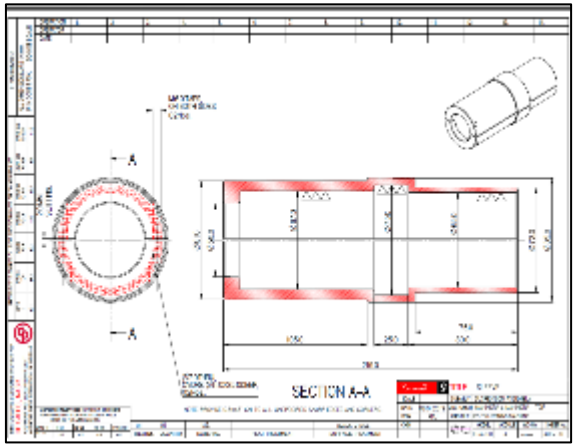
Corrective Action :

Metal sleeve for the protection of DC tool transducer wire found pinch out



Metal sleeve

Metal sleeve fitted on DC tool & attached to holder assembly



Metal sleeve drawing

6. Implement the corrective measures

6.1 Actions Decided

S N	Action Taken	Testing Method	Results Before Action	Results After Action
1	DC Tool transducer wire protection metal sleeve provided.	Visual inspection	Torque value was not reach 230NM for most of cycle	DC tool torque value reach 230NM for all cycle.

Metal sleeve is fitted

Sr. No	Status	ID	Tool	Torque	Angle	Stop source	Date	Result
1	OK	598	2	230.1	110	Target reached	12-May-16	1
2	NOK	597	2	79.32	291	Trigger released	12-May-16	1
3	NOK	596	2	41.74	1156	Trigger released	12-May-16	1
4	OK	595	2	230.6	101	Target reached	12-May-16	1
5	OK	594	2	230.9	117	Target reached	12-May-16	1
6	NOK	593	2	160.5	14.6	Trigger released	12-May-16	1
7	OK	592	2	230.7	175	Target reached	12-May-16	1
8	OK	591	2	230.3	105	Target reached	12-May-16	1
9	NOK	590	2	100.8	1366	Trigger released	12-May-16	1
10	OK	589	2	230.7	126	Target reached	12-May-16	1
11	OK	588	2	230.7	18.5	Target reached	12-May-16	1
12	NOK	587	2	202	9.7	Trigger released	12-May-16	1
13	NOK	586	2	210.5	0	Trigger released	12-May-16	1
14	NOK	585	2	207.4	108	Trigger released	12-May-16	1
15	OK	584	2	230.1	127	Target reached	12-May-16	1
16	OK	583	2	230.1	109	Target reached	12-May-16	1
17	NOK	582	2	12.49	151	Trigger released	12-May-16	1
18	OK	581	2	230.1	123	Target reached	12-May-16	1
19	OK	580	2	230.4	153	Target reached	12-May-16	1
20	NOK	579	2	98.27	1638	Trigger released	12-May-16	1
21	OK	578	2	230.2	128	Target reached	12-May-16	1
22	OK	577	2	230.6	125	Target reached	12-May-16	1
23	NOK	576	2	90.88	1707	Trigger released	12-May-16	1
24	OK	575	2	231	132	Target reached	12-May-16	1
25	OK	574	2	97.07	0	Target reached	12-May-16	1

Before

Sr. No	Status	ID	Tool	Torque	Angle	Stop source	Date	Result	TU
1	OK	15253	1	40.25	107	Trigger released	10-Aug-16	1	2
2	OK	15252	2	230.9	107	Target reached	10-Aug-16	1	2
3	OK	15251	1	230.7	86.1	Target reached	10-Aug-16	1	2
4	OK	15250	2	37.09	82.8	Trigger released	10-Aug-16	1	2
5	OK	15249	1	230.8	115	Target reached	10-Aug-16	1	2
6	OK	15248	2	230	103	Target reached	10-Aug-16	1	2
7	OK	15247	1	230.8	105	Target reached	10-Aug-16	1	2
8	OK	15246	2	230.2	153	Target reached	10-Aug-16	1	2
9	OK	15245	1	230.7	132	Target reached	10-Aug-16	1	2
10	OK	15244	2	230.4	125	Target reached	10-Aug-16	1	2
11	OK	15243	1	230.3	132	Target reached	10-Aug-16	1	2
12	OK	15242	2	230.1	105	Target reached	10-Aug-16	1	2
13	OK	15241	1	230.8	109	Target reached	10-Aug-16	1	2
14	OK	15240	2	230.7	106	Target reached	10-Aug-16	1	2
15	OK	15239	1	230.9	119	Target reached	10-Aug-16	1	2
16	OK	15238	2	230	107	Target reached	10-Aug-16	1	2
17	OK	15237	1	230.7	154	Target reached	10-Aug-16	1	2
18	OK	15236	2	230.5	133	Target reached	10-Aug-16	1	2
19	OK	15235	1	230.3	87.2	Target reached	10-Aug-16	1	2
20	OK	15234	2	230.1	108	Target reached	10-Aug-16	1	2
21	OK	15233	1	230.3	115	Target reached	10-Aug-16	1	2
22	OK	15232	2	230.4	86.8	Target reached	10-Aug-16	1	2
23	OK	15231	1	230.8	115	Target reached	10-Aug-16	1	2
24	OK	15230	2	109.3	130	Trigger released	10-Aug-16	1	2

After

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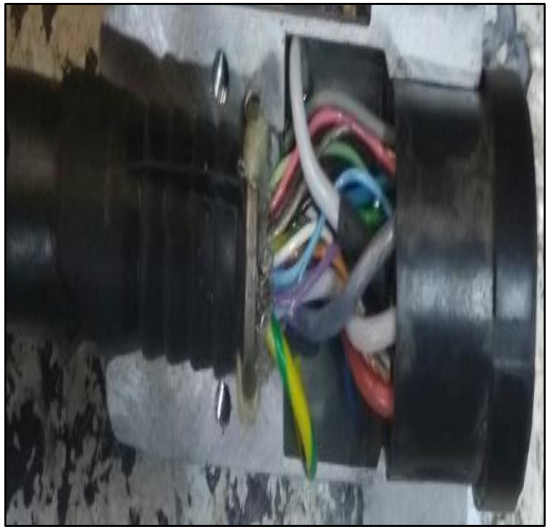
6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
3	Communication/ supply cable connector faulty.	Communication cable contact pin replaced with proper soldering done to it with locking provided.	3.08.2019	Jitendra	Closed



Transducer Wire pinch

Connector soldering to contact pin

Connector provide with locking

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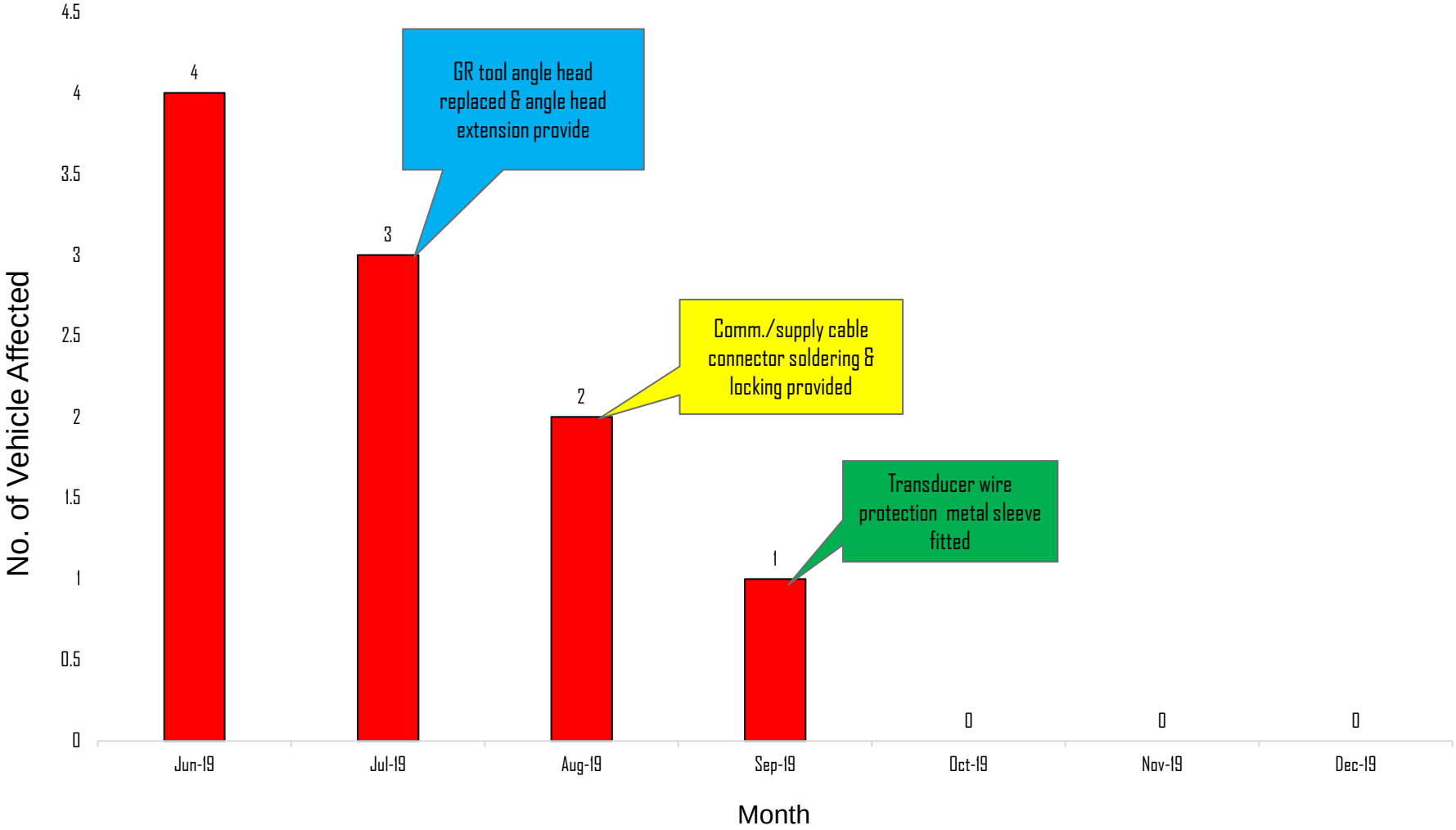
7. Check & Confirm the Results & Effects

7.1 Verify Results:



7. Check & confirm the results & effects

7.3 Full Scale Implementation plan:

Sr. No.	Action Taken	Actual date	Responsibility	Status
1.	DC tool angle head replaced.	11.08.2019	Jitendra	Closed
2.	DC tool angle head extension provided.	24.08.2019	Jitendra	Closed
3.	Transducer wire protection metal sleeve provided.	07.09.2019	Jitendra	Closed
4	Communication cable contact pin replaced with proper soldering done to it & Locking provided.	03.08.2019	Jitendra	Closed
5	Stack light provided to confirm the torque delivered.	21.09.2019	Jitendra	Closed
6	PM check sheet is updated.	28.09.2019	Jitendra/Hari	Closed
7	Metal sleeve drawing is updated	30.09.2019	Jitendra	Closed

Check & confirm the results & effects (Cont……)

7.4 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of DC tool failure issue.
2) Rework cost is reduced by approx.Rs.22560/Annum.
3) Spare material cost reduced by approx. Rs.121526.11/Annum.

In tangible Benefits :

1.Operator Moral enhancement.

Mon 11/18/2019 7:49 AM
DHANOKAR RAJESH
Re:cost saving approval
To BHAMARE JITENDRA; DHANOKAR RAJESH

okay approved

Sent from my Redmi Note 4X
On BHAMARE JITENDRA <BHAMARE.JITENDRA@mahindra.com>, 17-Nov-2019 10:22 AM wrote:
Dear Sir,
Sending here with Cost Saving data of Rework at Body mounting bolt tightening with Cleco DC tool for your approval

Rework cost estimation for Cleco DC Tool Failure issues

Description	Units	
Rework time(Approx)	Min	240
Avg Concerns/Day	Nos	0.13
Total Required Rework time per day	Min	31.2
Effective working time	Min	465
Operator required/day	Nos	0.06
Approx Operator cost/ day	Rs	1360
Operator cost / concern	Rs	75.2
Avg no. of working days	days	25
Operator cost /month	Rs	1880
Operator cost/year	**Rs**	22560

8. Standardization

8.1 Updation of documents:

SN	Document	Actual date	Responsibility	Status
1	PM check sheet is updated.	28.09.2019	Jitendra Bhamare	Closed
2	Metal sleeve drawing is updated	30.09.2019	Jitendra Bhamare	Closed

8.2 Training:

- Training given to stage associate that for all model there is torque range sheet attached for the reference.
- Also for camber bolt degree setting training given at IFS stage.

8.3 Additional Audit & check

- Stack light provided to confirm the torque value 230NM is delivered for both DC tool for stage associate easy operation.
- PM check sheet updated for monitoring the actions

8. Standardization

8.1 Updation of documents:

SN	Document	Actual date	Responsibility	Status
1	PM check sheet is updated.	28.09.2019	Jitendra Bhamare	Closed

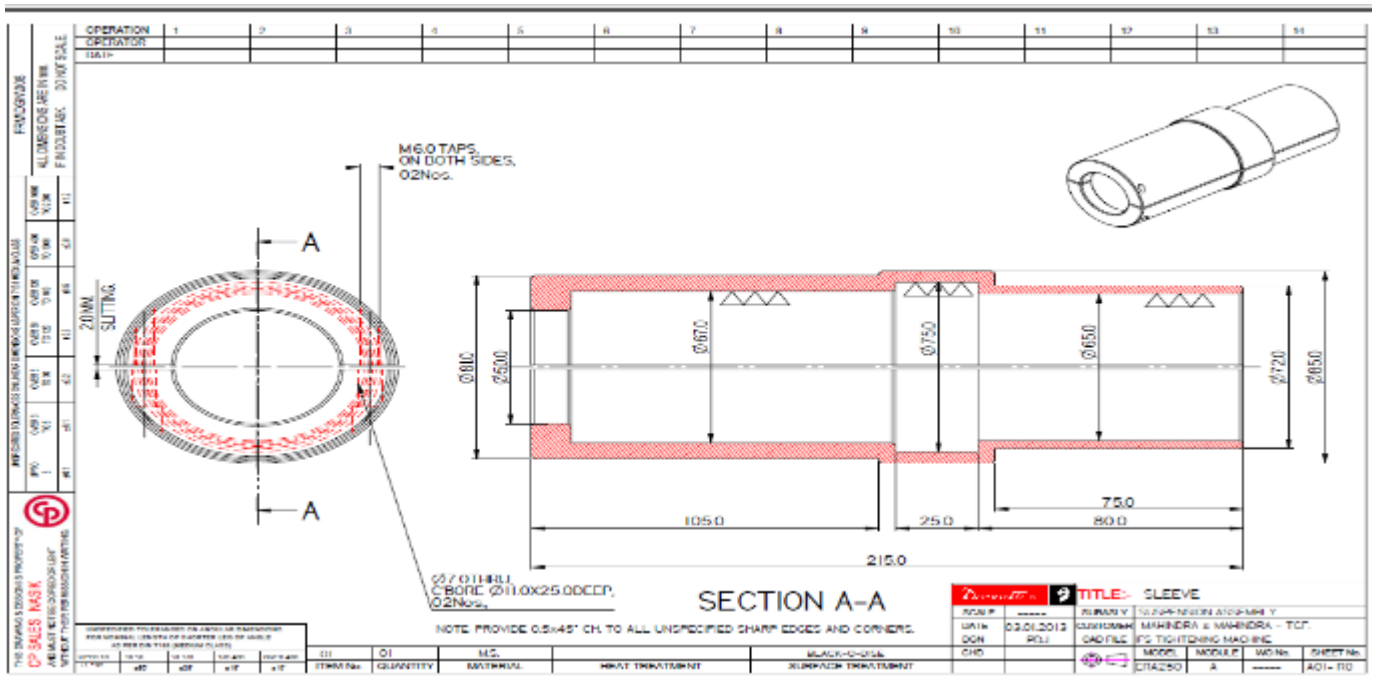
OFF line PM Inspection Sheet (Scorpio TCF Maintenance)					Frequency :Monthly	
Area :- Chasssis Line		Name of Machine :- GR tool .....with Articulate arm (Saturday)			PM Date :	
Sr. No	Item (What)	Activity	Method (How)	Required Value	C/NC	Remark
1	Controller	Check	physical	All cable connected to the controller are tightened properly	○	
		Check	physical	Temp.cap & Emm cap on controller must be tighttned	○	
2	Scoket Selection Tray **(For IFS Stage only)**	Check	Physical	Cable connections ok	○	
		Check	physical	LED operation as per scoket selection ok	○	
		Check	physical	Socket tray mounting Ok	○	
3	All area	Clean	Physical with cloth & brush	No Dust, no oil deposits	○	
4	Tool arm	Noise	Listen	There should not be any abnormal noise during running. Clean rusty part at joining, locking pin	○	
5	Rail	Check	Physical	Free movement of gun for smoothness in Horizontal Direction	○	
		Check	visual	All Rail Mounting Bolt/clamp should be tightened as per **match marking**	○	
6	FRL Checking	Check	Visual	Filter cleaning for No clogging	○	
		Check	visual	Oil level should not be below "min level"	○	
7	Pneumatic pipes	Check	Physical	Check proper routine of pipe & without leakage & Pinching	○	
8	Gr tool Cable	Check	Physical	GR tool cable routine should be proper & no any sharp bend / kink for cable.	○	
9	Tool Holding assly & Arm	Check	visual	Check indexing of assly & bearing, clean & lubricate for smothness operation	○	
				Check the fastners for tighttness as per **match marking**	○	
10	Pneumatic Cylinder	Check	physical	Up / Down movement ok. Mounting ok, Locking pin with Tighten Lock nut for arm free movement	○	
11	Flow regulator	Check	physical	Set regulator for smooth movement	○	
12	Tool arm	Lubricate	Physical	All joints lubricate to minimize fractional loss	○	
13	PY limit switch	check	physical	Limit switch mounting ok. Switch wiring connection tightning ok. Wire routine ok. Switch functioning ok.(Line stop if L/S gets sense)	○	
14	Pokayoke check (For multilink GR tool)	check	physical	GR Tool should operate only when rear compression is in operate condition with Axel.	○	
15	Poka yoke - Torque selection	check	physical	GR Tool Torque should get select by scanner .	○	

Rev No.	Revised on	Summery of Revision (Not". No)	Reason of Revision (B/D)	Kaizen (If any)
1				
Checked By :			Verified By :	

8. Standardization

8.1 Updation of documents:

SN	Document	Actual date	Responsibility	Status
2	Metal sleeve drawing is updated.	30.09.2019	Jitendra Bhamare	Closed



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8. Standardization

8.2 Training:

- Training given to stage associate that for all model there is torque range sheet attached for the reference.
- Also for camber bolt degree setting training given at IFS stage.
- OPL given for DC Tool parking in parking stand after cycle completion.

OPL - One Point Lesson

Mahindra

Does:
1.GR Tool should park in parking stand.
2.GR tool cable should operate carefully.

Don't:
1. Do not release GR tool in open space(Other than parking area).
2. Do not pull GR tool with help of cable.

OPL given to associates

8. Standardization

8.3 Additional Audit & check

a) Stack light provided to confirm the torque value 230NM is delivered for both DC tool for stage associate easy operation.

b) PM check sheet updated for monitoring the actions

OFF line PM Inspection Sheet (Scorpio TCF Maintenance)					Frequency :Monthly	
Area :- Chasssis Line		Name of Machine :- GR tool .....with Articulate arm (Saturday)			PM Date :	
Sr. No	Item (What)	Activity	Method (How)	Required Value	C/NC	Remark
1	Controller	Check	physical	All cable connected to the controller are tightened properly	○	
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		Check	physical	LED operation as per scoket selection ok	○	
		Check	physical	Socket tray mounting Ok	○	
3	All area	Clean	Physical with cloth & brush	No Dust, no oil deposits	○	
4	Tool arm	Noise	Listen	There should not be any abnormal noise during running. Clean rusty part at joining, locking pin	○	
5	Rail	Check	Physical	Free movement of gun for smoothness in Hortzontal Direction	○	
		Check	visual	All Rail Mounting Bolt/clamp should be tightened as per **match marking**	○	
6	FRL Checking	Check	Visual	Filter cleaning for No clogging	○	
		Check	visual	Oil level should not be below "min level"	○	
7	Pneumatic pipes	Check	Physical	Check proper routine of pipe & without leakage & Pinching	○	
8	Gr tool Cable	Check	Physical	GR tool cable routine should be proper & no any sharp bend / kink for cable.	○	
9	Tool Holding assly & Arm	Check	visual	Check indexing of assly & bearing, clean & lubricate for smothness operation	○	
				Check the fastners for tighttness as per **match marking**	○	
10	Pneumatic Cylinder	Check	physical	Up / Dawn movement ok. Mounting ok, Locking pin with Tighten Lock nut for arm free movement	○	
11	Flow regulator	Check	physical	Set regulator for smooth movement	○	
12	Tool arm	Lubricate	Physical	All joints lubricate to minimize fractional loss	○	
13	PY limit switch	check	physical	Limit switch mounting ok. Switch wiring connection tightning ok. Wire routine ok. Switch functioning ok.(Line stop if L/S gets sense)	○	
14	Pokayoke check (For multilink GR tool)	check	physical	GR Tool should operate only when rear compression is in operate condition with Axel.	○	
15	Poka yoke - Torque selection	check	physical	GR Tool Torque should get select by scanner .	○	
Rev No.	Revised on	Summery of Revision (Not". No)		Reason of Revision (B/D)		Kaizen (If any)
1						
Checked By :					Verified By :	

PM check sheet updated

9. Summaries & Plan future actions

9.1 Conclusion:

Concern eliminated with replacement of angle head, provision of angle head extension and metal sleeve provided for protection of transducer wire.

9.2 Horizontal Deployment:

Ele DC tool at multilink tightening stage, Wheel alignment and chassis offline stage initiated.

9.3 Lessons Learnt:

Things Gone Wrong :

1.DC tool failure reactive Approach to resolve the issue.
2.Controller comm. With DC tool failure was considered & but was found Ok

Things Gone Right :

1. DC tool transducer consider as malfunction and found its wire Pinch.
2. DC tool connector of comm. Cable having more tension was identified as a cause & given good results.

Thank you

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